

Fig. 6: Fully saturated aprotic lithium-air cathode

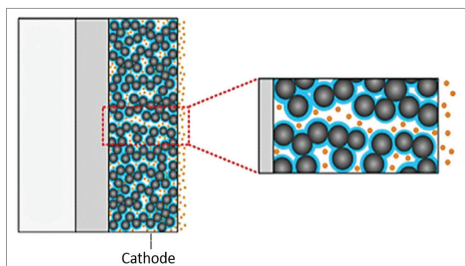


Fig. 7: Fully wetted and partially saturated aprotic lithium-air cathode

new philosophy has now been developed that uses lithium as a soluble catalyst and the resulting battery can run stably for more than 900 cycles.

Cathode materials should have a durable porous structure, enabling it to store discharge products and provide channels for oxygen diffusion. They should have high electrolyte wettability for ionic transfer during charging and discharging. Cathodes should have the ability to accelerate the kinetics of oxygen reduction/evolution reaction (ORR/OER). In non-aqueous lithium-air batteries carbon materials are generally used to fabricate a porous cathode and they also can work as catalyst towards ORR/OER.

Carbon material with surface areas and larger number of mesopores could deliver higher discharge capacities. Functional carbon materials that are used as cathode substrate are generally graphene, carbon nanotubes (CNT), mesoporous carbon with pores having diameter between 2 and 50nm mixed with cobalt and manganese as catalyst, which enhances reaction kinetics and increases cathode's specific capacity. They are quite effective exhibiting discharge capacities ranging from 7000 to 8700mAh per gm C.

In conventional aprotic lithium-air batteries, the porous carbon composite air electrode (cathode) is fully saturated with the electrolyte, as shown in Fig. 6, which retards the transport rate of oxygen in the electrolyte to the reaction sites and the oxygen concentration at reaction site is very much decreased. Oxygen is first dissolved into the electrolyte at the oxygen/electrolyte interface and is then transported to the reaction sites. The insufficient supply of oxygen near the separator side limits the chemical reaction, leading to a lower power density output.

A possible remedy is to replace the fully saturated cathode into a partially saturated one by special chemical treatment, as shown in Fig. 7. In this concept, distribution of electrolyte inside the cathode pores due to partial saturation enables creation of uniform concentration distribution of both oxygen and lithium ions on the entire cathode surface. Gaseous oxygen can rapidly penetrate the pores and can be quickly transported to the reaction sites, leading to improved battery performance.

The separator in a lithium-air battery system is typically a porous polymer membrane that is wetted by the liquid electrolyte and located between cathode and anode, as shown in Fig. 8. The membrane separator is a key component in a

liquid electrolyte battery for electrical separating the cathode and anode, meanwhile ensuring ionic transport between them.

In lithium-air batteries, the migration of moisture and oxygen from the air cathode to the lithium anode leads to unstable cycling. Therefore, ion-selective separators are required. The membrane materials are Nafion, LISICON, polymer of intrinsic microporosity, polyurethane, graphene oxide, polyethelene separator treated with polydopamine, etc. They allow the electrochemical reactions at both lithium metal anode and the air cathode to take place due to high lithium-ion conductivity. Further, they protect the anode from corrosion as moisture and oxygen are unable to penetrate through the separator membrane.

Lithium dendrite formation on lithium metal anode during charging/discharging in lithium-air batteries is a one way process and is quite harmful, causing performance decay. The above separators suppress lithium dendrite formation and show improved thermal stability against dimensional shrinkage at elevated temperature.

Aqueous electrolyte battery

An aqueous lithium-air battery consists of a lithium metal anode, an aqueous electrolyte, and a porous carbon cathode impregnated with silver. The aqueous electrolyte combines lithium salts dissolved in water. Here the problem of cathode clogging is solved because the reaction products are water-soluble.

This design has a higher practical discharge potential of 3.84V than aprotic type, which has cell voltage of about 2.92V. The energy density of aqueous system is about 30% lower than that of nonaqueous system. As lithium metal reacts violently with water, it is necessary to have water-stable lithium electrodes (WSLE) to obtain a practical aqueous lithium-air secondary battery that can survive lithium stripping for a long life span.

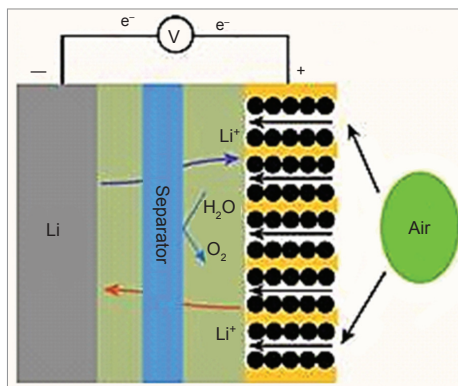


Fig. 8: Separator in a lithium-air electrolyte battery